Change one heuristic to an overestimate

The manual specifically asks to change only $h(C)$.

Heuristic experiment

Original value

h(C) = 2

Modified value

h(C) = 10

C's true remaining cost is 3, because the only route from C to G costs 3.

$$h(C) = 10 > 3$$

The modified heuristic overestimates the remaining cost by $10 - 3 = 7$.

3.1 What happens after the change?

After expanding S and A, the frontier contains C and B.

For C:

$$g(C) = 1 + 2 = 3$$

[f(C)=3+10=\boxed{13} \]

For B:

$$g(B) = 0 + 4 = 4$$

\[f(B)=4+4=\boxed{8} \]

Since $8 < 13$, A* selects B instead of C.

B generates G:

$$g(G) = 4 + 5 = 9$$

\[f(G)=9+0=\boxed{9} \]

Now G has a lower priority than C, because $9 < 13$. A* selects G and returns the route through B before examining C.

3.2 Before-and-after results

Measurement	Original heuristic	Modified $h(C) = 10$
Returned path	S → A → C → G	S → B → G
Independently summed cost	$1 + 2 + 3 = 6$	$4 + 5 = 9$
Processed trace	S, A, C, G	S, A, B, G
Expanded nodes	S, A, C	S, A, B
Expansion count	3	3

Result: The returned path and cost change, but the expansion count stays at 3.

3.3 Why does overestimating matter?

The original heuristic is admissible on this graph: its values do not exceed the true remaining costs. Setting $h(C) = 10$ makes C look much more expensive than it actually is, so the algorithm returns the more expensive route through B.

This experiment shows that an overestimating heuristic can remove A's optimality guarantee*. It does not mean every overestimate will change every result; an unchanged result in a different experiment would not prove that the heuristic is safe.